

TerraTwin

Knowledge Base — 500+ Authentic Public Sources

Clickable references, organized by purpose (Part A: AI/Policy · Part B: Agricultural & Technical)

Submitted to: ITU AI Readiness Hackathon — Kingdom of Saudi Arabia 2026

Wanas Zakri — Management Information Systems, Jazan University

All 505 sources from the original submission are preserved verbatim, now presented as clickable hyperlinks in a structured table per section. Reference numbers (#1–#505) are unchanged for traceability. Nothing was removed, added, or reworded — only the link format and layout changed.

Ctrl/Cmd+click any link in the 'Link' column to open the source directly. Links were compiled by the author; re-verify shortly before final submission as institutional sites occasionally restructure.

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PART A — AI Readiness Policy, Standards, Governance & Institutional Sources

TerraTwin's direct contribution to the ITU AI Readiness Knowledge Base: ITU-T ML/AI standards, Saudi national AI/data/water policy, data protection & AI governance, and the UN SDG framework. 83 of the 505 total sources.

A.1.1 Core ITU-T Recommendations

#	Source	Context	Provided by	Link
1	ITU-T Y.3172	ML framework for future networks · Core standard — TerraTwin pipeline	ITU-T SG13	itu.int/rec/T-REC-Y.3172/en
2	ITU-T Y.3170	Requirements for ML in future networks · Requirements underpinning Y.3172	ITU-T SG13	itu.int/rec/T-REC-Y.3170/en
3	ITU-T Y.3176	ML marketplace framework	ITU-T SG13	itu.int/rec/T-REC-Y.3176/en
4	ITU-T Y.3173	Framework for evaluating ML models	ITU-T SG13	itu.int/rec/T-REC-Y.3173/en
5	ITU-T Y.3174	ML sandbox for future networks · ML Sandbox standard	ITU-T SG13	itu.int/rec/T-REC-Y.3174/en
6	ITU-T Y.3100	Terms and definitions for IMT-2020	ITU-T	itu.int/rec/T-REC-Y.3100/en
7	ITU-T Y.3104	Architecture of IMT-2020 networks	ITU-T	itu.int/rec/T-REC-Y.3104/en
8	ITU-T Y.3111	IMT-2020 network management framework	ITU-T	itu.int/rec/T-REC-Y.3111/en
9	ITU-T Y.3650	Big data framework	ITU-T	itu.int/rec/T-REC-Y.3650/en
10	ITU-T Y Series Recommendations	full list	ITU-T	itu.int/rec/T-REC-Y/en

A.1.2 AI for Good & Hackathon

#	Source	Context	Provided by	Link
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11	AI for Good platform	Hackathon organiser	ITU / UN	aiforgood.itu.int
12	AI Readiness Hackathon	Kingdom of Saudi Arabia · Official event page	AI for Good	aiforgood.itu.int/event/ai-readiness-hackathon-kingdom-of-saudi-arabia
13	AI Readiness Hackathon	Kenya (first edition)	AI for Good	aiforgood.itu.int/event/ai-readiness-hackathon-kenya/
14	ITU AI-RE Toolkit (GitHub)	Knowledge base target repository	ITU	github.com/CrashingGuru/ITUAIReadiness
15	AI for Good Neural Network		ITU	aiforgood.itu.int/about-ai-for-good/
16	ITU Publications	AI and emerging technologies	ITU	itu.int/en/publications/Pages/default.aspx
17	ITU-T Study Group 13	Future networks and AI	ITU-T	itu.int/en/ITU-T/studygroups/2022-2024/13/Pages/default.aspx
18	ITU-T Focus Group on ML for 5G		ITU-T	itu.int/en/ITU-T/focusgroups/ml5g/Pages/default.aspx
19	Government AI Readiness Index		Oxford Insights / IDRC	oxfordinsights.com/ai-readiness/ai-readiness-index/
20	UNESCO AI Readiness Assessment Methodology		UNESCO	unesco.org/en/artificial-intelligence

A.1.3 Related International Standards

#	Source	Context	Provided by	Link
21	ISO/IEC 23053	AI/ML framework	ISO/IEC	iso.org/standard/7443

				8.html
22	ISO/IEC 42001	AI management system	ISO/IEC	iso.org/standard/81230.html
23	IEEE 7000	Ethical design of autonomous systems	IEEE	standards.ieee.org/ieee/7000/6781/
24	OECD AI Principles		OECD	oecd.ai/en/ai-principles
25	EU AI Act	official text	European Union	artificialintelligenceact.eu

A.2.1 National Strategy & Vision

#	Source	Context	Provided by	Link
26	Saudi Vision 2030	official portal	Government of KSA	vision2030.gov.sa
27	Saudi Vision 2030	delivery dashboard	Government of KSA	vision2030.gov.sa/v2030/overview/
28	Saudi Vision 2030	environmental sustainability	Government of KSA	vision2030.gov.sa/v2030/a-vibrant-society/a-fulfilling-and-healthy-lif
29	National Strategy for Data and AI (NSDAI)		SDAIA	sdaia.gov.sa/en/SDAIA/SdaiaStrategies/Pages/NationalStrategyForDataAnd
30	SDAIA	about	SDAIA	sdaia.gov.sa/en/SDAIA/about/Pages/AboutSDAIA.aspx
31	SDAIA	regulations and policies · AI governance	SDAIA	sdaia.gov.sa/en/SDAIA/about/Pages/RegulationsAndPolicies.aspx

32	Personal Data Protection Law (PDPL)	Data protection basis	SDAIA	sdaia.gov.sa/en/Research/Pages/DataProtection.aspx
33	SDAIA	AI ethics principles	SDAIA	sdaia.gov.sa/en/SDAIA/about/Pages/RegulationsAndPolicies.aspx
34	National Transformation Program (NTP)		Government of KSA	vision2030.gov.sa/v2030/vrps/ntp/
35	Saudi Green Initiative		Saudi Green Initiative	saudigreeninitiative.com
36	Saudi Green Initiative	targets	Sgi	saudigreeninitiative.com/targets/
37	National Water Strategy		MEWA	mewa.gov.sa/en/
38	National Environment Strategy		Government of KSA	my.gov.sa/

A.2.2 MEWA — Agriculture & Water

#	Source	Context	Provided by	Link
39	MEWA	official portal	Ministry of Environment, Water and Agriculture	mewa.gov.sa/en/
40	MEWA	agriculture services	MEWA	mewa.gov.sa/en/InformationCenter/DocsCenter/Pages/default.aspx
41	MEWA	Naama e-services platform	MEWA	mewa.gov.sa/en/eServices/Pages/default.aspx
42	MEWA	Innovation Platform · Smart sensors, water analytics	MEWA	mewa.gov.sa/en/Ministry/AboutMinistry/Sharing/InnovationPlatform

				m/Pages
43	MEWA	water sector	MEWA	mewa.gov.sa/en/Ministry/Ministry/Pages/Water.aspx
44	MEWA	environment sector	MEWA	mewa.gov.sa/en/Ministry/Ministry/Pages/Environment.aspx
45	Organic farming in Saudi Arabia		MEWA	mewa.gov.sa/en/
46	Agricultural Development Fund (ADF)		ADF	adf.gov.sa/en
47	ADF	loans and programmes	ADF	adf.gov.sa/en/products
48	National Center for Vegetation Cover (NCVC)		NCVC	ncvc.gov.sa/en/
49	NCVC	desertification control	NCVC	ncvc.gov.sa/en/
50	Saudi Wildlife Authority (now NCWCD)		NCWCD	ncw.gov.sa/En/Pages/default.aspx
51	Weqaa	plant pest and disease control	Weqaa	weqaa.gov.sa/
52	Saudi Food and Drug Authority (SFDA)		SFDA	sfda.gov.sa/en
53	SFDA	food safety standards	SFDA	sfda.gov.sa/en/food

A.2.3 Statistics & Data

#	Source	Context	Provided by	Link
54	General Authority for Statistics (GASTAT)		GASTAT	stats.gov.sa/en
55	GASTAT	agricultural statistics	GASTAT	stats.gov.sa/en

56	GASTAT	Comprehensive Agricultural Survey	GASTAT	stats.gov.sa/en/w/gastat-launches-comprehensive-agricultural-survey
57	GASTAT	environmental statistics	GASTAT	stats.gov.sa/en
58	GASTAT	open data portal	GASTAT	data.stats.gov.sa/
59	Saudi Central Bank	economic data	SAMA	sama.gov.sa/en-us/pages/default.aspx

A.3 Data Protection & AI Governance

#	Source	Context	Provided by	Link
348	Saudi PDPL	full text	SDAIA	sdaia.gov.sa/en/Research/Pages/DataProtection.aspx
349	SDAIA	AI ethics principles	SDAIA	sdaia.gov.sa/en/SDAIA/about/Pages/RegulationsAndPolicies.aspx
350	GDPR	EU data protection	EU	gdpr.eu
351	OECD	AI policy observatory	OECD	oecd.ai
352	Global Partnership on AI (GPAI)		GPAI	gpai.ai
353	Partnership on AI		PAI	partnershiponai.org
354	AI Index	Stanford HAI	Stanford	aiindex.stanford.edu
355	Responsible AI	Microsoft	Microsoft	microsoft.com/en-us/ai/responsible-ai
356	Google AI Principles		Google	ai.google/responsibility/principles/
357	Ethics guidelines for	EU	European	digital-strategy.ec.europa.eu/

	trustworthy AI		Commission	en
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A.4 Sustainable Development & SDGs (UN policy framework)

#	Source	Context	Provided by	Link
439	SDG 2	Zero Hunger	UN	sdgs.un.org/goals/goal2
440	SDG 6	Clean Water and Sanitation	UN	sdgs.un.org/goals/goal6
441	SDG 12	Responsible Consumption	UN	sdgs.un.org/goals/goal12
442	SDG 13	Climate Action	UN	sdgs.un.org/goals/goal13
443	SDG 15	Life on Land	UN	sdgs.un.org/goals/goal15
444	FAO and the SDGs		FAO	fao.org/sustainable-development-goals/en/
445	Voluntary National Review	KSA	UN	sustainabledevelopment.un.org
446	Saudi Arabia SDG dashboard		Sachs et al.	dashboards.sdindex.org
447	World Resources Institute	food	WRI	wri.org/food
448	IRENA	renewable energy for agriculture	IRENA	irena.org

A.5 Additional AI/Standards-Adjacent Institutional References

#	Source	Context	Provided by	Link
493	Saudi Authority for Data and AI	careers	SDAIA	sdaia.gov.sa/en/

494	IMT-2020 network architecture	ITU	ITU-T	itu.int/rec/T-REC-Y/en
495	IEEE	communications standards	IEEE	ieee.org
496	ITU	publications on AI	ITU	itu.int/en/publications/Pages/default.aspx

PART B — Agricultural Domain Knowledge, Research & Technical/Development References

Grounds TerraTwin's Farmy assistant (crop, soil, water, pest, livestock, regional knowledge) and documents the technical/development tooling used to build the prototype. 422 of the 505 total sources — authentic public sources, but not themselves national AI policy papers, which is why they are kept separate from Part A.

B.1 FAO & Global Agriculture Databases

#	Source	Context	Provided by	Link
60	FAO	official site	FAO	fao.org
61	FAOSTAT	statistical database	FAO	fao.org/faostat/en/
62	FAO	country profile: Saudi Arabia	FAO	fao.org/countryprofiles/index/en/?iso3=SAU
63	FAO AQUASTAT	water resources data	FAO	fao.org/aquastat/en/
64	FAO	crop water information	FAO	fao.org/land-water/databases-and-software/crop-information/en/
65	FAO	irrigation and drainage	FAO	fao.org/land-water/water/en/
66	FAO	soil portal	FAO	fao.org/soils-portal/en/
67	FAO	world soil resources	FAO	fao.org/soils-portal/data-hub/soil-maps-and-databases/en/
68	FAO	integrated pest management	FAO	fao.org/pest-and-pesticide-management/ipm/en/
69	FAO	plant production and protection	FAO	fao.org/agriculture/crops/en/
70	FAO	climate-smart agriculture	FAO	fao.org/climate-smart-agriculture/en/

71	FAO	climate-smart agriculture sourcebook	FAO	fao.org/climate-smart-agriculture-sourcebook/en/
72	FAO	sustainable food and agriculture	FAO	fao.org/sustainability/en/
73	FAO	food loss and waste	FAO	fao.org/food-loss-and-food-waste/en/
74	FAO	agroecology knowledge hub	FAO	fao.org/agroecology/en/
75	FAO	digital agriculture	FAO	fao.org/digital-agriculture/en/
76	FAO	global soil partnership	FAO	fao.org/global-soil-partnership/en/
77	FAO	world food situation	FAO	fao.org/worldfoodsituation/en/
78	FAO	food price index	FAO	fao.org/worldfoodsituation/foodpricesindex/en/
79	FAO	agricultural outlook	FAO/OECD	fao.org/publications/oecd-fao-agricultural-outlook/en/
80	FAO	sustainable land management	FAO	fao.org/land-water/land/en/
81	FAO	livestock	FAO	fao.org/animal-production/en/
82	FAO	fisheries and aquaculture	FAO	fao.org/fishery/en
83	FAO	codex alimentarius (food standards)	FAO/WHO	fao.org/fao-who-codexalimentarius/en/
84	FAO	GIEWS food security monitoring	FAO	fao.org/giews/en/

85	FAO	investment in agriculture	FAO	fao.org/investment-in-agriculture/en/
86	FAO	water productivity	FAO	fao.org/land-water/water/en/
87	FAO	crop ecology and land management	FAO	fao.org/agriculture/crops/en/
88	FAO	biotechnologies in agriculture	FAO	fao.org/biotechnology/en/
89	IFAD	smallholder agriculture	IFAD	ifad.org
90	WFP	food security analysis	WFP	wfp.org
91	UNEP	sustainable agriculture	UNEP	unep.org
92	UNESCO	intangible cultural heritage · Khawlani coffee inscription	UNESCO	ich.unesco.org
93	UN Water	water scarcity	UN Water	unwater.org
94	UNCCD	desertification	UNCCD	unccd.int
95	World Bank	agriculture data	World Bank	data.worldbank.org/topic/agriculture-and-rural-development
96	World Bank	water resources	World Bank	worldbank.org/en/topic/water

B.2 Research Institutions

#	Source	Context	Provided by	Link
97	ICBA	International Center for Biosaline Agriculture · Saline agriculture research	ICBA	biosaline.org
98	ICBA	research programmes	ICBA	biosaline.org/research

99	ICBA	halophyte crops	ICBA	biosaline.org
100	ICBA	quinoa research	ICBA	biosaline.org
101	KAUST	plant science · Desert agriculture research	KAUST	kaust.edu.sa
102	KAUST	Center for Desert Agriculture	KAUST	kaust.edu.sa
103	KAUST	Red Sea Research Center	KAUST	kaust.edu.sa
104	King Saud University	College of Agriculture	KSU	cfa.ksu.edu.sa/en
105	King Faisal University	Agriculture College	KFU	kfu.edu.sa/en
106	Jazan University		JU	jazanu.edu.sa/en
107	UAE University	College of Agriculture	UAEU	uae.ac.ae/en/cas/
108	Masdar Institute / Khalifa University		Khalifa University	ku.ac.ae
109	KISR	Kuwait Institute for Scientific Research	KISR	kisir.edu.kw/en/
110	AOAD	Arab Organization for Agricultural Development	AOAD	aoad.org
111	CGIAR	research for food security	CGIAR	cgiar.org
112	ICARDA	dry-area agriculture research · Dry-area cereals and legumes	CGIAR	icarda.org
113	ICARDA	barley improvement	ICARDA	icarda.org
114	ICARDA	wheat improvement	ICARDA	icarda.org
115	ICRISAT	semi-arid tropics research	CGIAR	icrisat.org
116	CIMMYT	maize and wheat	CGIAR	cimmyt.org
117	IRRI	rice research	CGIAR	irri.org
118	IITA	tropical agriculture	CGIAR	iita.org

119	CIP	potato and sweet potato	CGIAR	cipotato.org
120	ICRAF	agroforestry	CGIAR	worldagroforestry.org
121	Bioversity International	agrobiodiversity	CGIAR	bioversityinternational.org
122	WorldFish	aquaculture and fisheries	CGIAR	worldfishcenter.org
123	IWMI	water management · Irrigation and water productivity	CGIAR	iwmi.cgiar.org
124	Wageningen University	agricultural sciences	WUR	wur.nl/en.htm
125	UC Davis	plant sciences	UC Davis	ucdavis.edu
126	Cornell University	College of Agriculture	Cornell	cals.cornell.edu
127	CIRAD	agricultural research for development	CIRAD	cirad.fr/en
128	INRAE	French agri research	INRAE	inrae.fr/en
129	Rothamsted Research	crop science	Rothamsted	rothamsted.ac.uk
130	CSIRO	agriculture and food	CSIRO	csiro.au/en/research/natural-environment/land/agriculture
131	CABI	agricultural biosecurity	CABI	cabi.org
132	CABI	Invasive Species Compendium	CABI	cabidigitallibrary.org/product/qi
133	Bioversity International	crop diversity	Alliance Bioversity-CIAT	alliancebioversityciat.org

B.3 Crop Science & Horticulture

#	Source	Context	Provided by	Link
134	Date palm cultivation	FAO	FAO	fao.org/3/y4360e/y43

				60e00.htm
135	Date palm genetic resources	Bioversity	Bioversity	bioversityinternational.org
136	Red palm weevil management	FAO	FAO	fao.org/red-palm-weevil/en/
137	Saudi date varieties database		MEWA	mewa.gov.sa/en/
138	King Faisal University	date palm research	KFU	kfu.edu.sa/en
139	Al-Ahsa World Heritage	date palm oasis	UNESCO	whc.unesco.org/en/list/1563/
140	Date palm products and processing		FAO	fao.org
141	International Date Palm Network		Bioversity	bioversityinternational.org
142	Dubas bug management		FAO	fao.org
143	Date palm tissue culture		KAUST	kaust.edu.sa
144	Khawlani coffee	UNESCO ICH inscription · Traditional knowledge recognition	UNESCO	ich.unesco.org
145	Coffee leaf rust	World Coffee Research	WCR	worldcoffeeresearch.org
146	World Coffee Research	varieties	WCR	varieties.worldcoffeeresearch.org
147	ICO	International Coffee Organization	ICO	ico.org
148	Arabica coffee agronomy		FAO	fao.org/agriculture/crops/en/
149	Shade-grown coffee systems		ICRAF	worldagroforestry.org
150	International Olive Council (IOC)		IOC	internationaloliveoil.org

151	IOC	olive oil standards	IOC	internationaloliveoil.org
152	Al-Jouf olive industry		MEWA / Al-Jouf	mewa.gov.sa/en/
153	Olive cultivation in arid conditions	FAO	FAO	fao.org
154	Olive oil quality assessment		IOC	internationaloliveoil.org
155	Citrus rootstock selection guide		UC Riverside	citrusvariety.ucr.edu
156	Citrus greening (HLB)	USDA	USDA	aphis.usda.gov/plant-pests-diseases/citrus
157	Citrus psyllid management		FAO	fao.org
158	FAO	citrus fruit statistics	FAO	fao.org/faostat/en/
159	Citrus varieties for hot climates		ICARDA	icarda.org
160	FAO	vegetable production guide	FAO	fao.org/agriculture/crops/en/
161	Greenhouse vegetable production	FAO	FAO	fao.org
162	Tomato production in protected cultivation		FAO	fao.org
163	Cucumber and cucurbit cultivation		AVRDC	avrdc.org
164	AVRDC	World Vegetable Center · Tropical vegetable research	WorldVeg	avrdc.org
165	Seed Alliance	vegetable varieties	Global Seed Alliance	worldveg.org
166	Pepper and chilli diversity		WorldVeg	avrdc.org
167	Okra	heat-tolerant vegetable	WorldVeg	avrdc.org

168	Potato cultivation	CIP	CIP	cipotato.org
169	Onion and garlic production	FAO	FAO	fao.org
170	Mango production guide	FAO	FAO	fao.org
171	Banana and plantain	Bioversity	Bioversity	bioversityinternational.org
172	Guava cultivation and post-harvest		FAO	fao.org
173	Papaya production		FAO	fao.org
174	Avocado cultivation	UC Davis	UC Davis	ucdavis.edu
175	Pineapple agronomy		FAO	fao.org
176	Tropical fruit network		TFNet	itfnet.org
177	Apple chill-hour requirements		Utah State University Extension	extension.usu.edu
178	Stone fruit cultivation guide		UC Davis	ucdavis.edu
179	Almond production		Almond Board of California	almonds.com
180	Pistachio cultivation		UC Davis	ucdavis.edu
181	Walnut production guide		UC Davis	ucdavis.edu
182	Grape and vine management		UC Davis Viticulture	ucdavis.edu
183	Pomegranate cultivation		UC Davis	ucdavis.edu
184	Fig production		UC Davis	ucdavis.edu
185	Low-chill fruit varieties		University of Florida	edis.ifas.ufl.edu
186	Strawberry production systems		UC Davis	ucdavis.edu

187	Chill-hour calculation methods		UC Davis	ucdavis.edu
188	Growing degree days for crop planning		NOAA	weather.gov
189	Cactus pear / prickly pear	FAO	FAO	fao.org/cactus-pear/en/
190	Wheat production	FAO	FAO	fao.org
191	Barley	salt and drought tolerant cereal	ICARDA	icarda.org
192	Sorghum production in dry areas		ICRISAT	icrisat.org
193	Millet	climate-resilient cereal	ICRISAT	icrisat.org
194	Rice production	IRRI	IRRI	irri.org
195	Maize production	CIMMYT	CIMMYT	cimmyt.org
196	Quinoa	salt-tolerant grain	FAO	fao.org/quinoa/en/
197	FAO	quinoa promotion	FAO	fao.org/quinoa/en/
198	Sesame production		FAO	fao.org
199	Cotton cultivation		ICAC	icac.org
200	Sugar beet production		FAO	fao.org
201	Alfalfa water use	Arizona	University of Arizona	extension.arizona.edu
202	Hydroponic green fodder systems		ICBA	biosaline.org
203	Halophyte forage crops	ICBA	ICBA	biosaline.org
204	Salicornia	seawater-irrigated crop	ICBA	biosaline.org
205	Cover crops and green manure	FAO	FAO	fao.org/conservation-agriculture/en/

206	Ghaf tree	native dryland species	IUCN	iucn.org
207	Acacia species in Arabia		IUCN	iucn.org
208	Neem	botanical pesticide source	NRC	nap.edu
209	Moringa	nutritious drought-tolerant tree	Moringa Association	moringanews.org
210	Sidr / jujube	Arabian native fruit tree	FAO	fao.org
211	Tamarisk	salt-tolerant windbreak	FAO	fao.org
212	Frankincense	Boswellia conservation	IUCN	iucn.org
213	Desert restoration and reforestation		UNCCD	unccd.int
214	Agroforestry in dry regions	ICRAF	ICRAF	worldagroforestry.org
215	Saudi juniper woodland conservation		NCVC	ncvc.gov.sa/en/

B.4 Soil Science & Management

#	Source	Context	Provided by	Link
216	FAO	world soil map	FAO	fao.org/soils-portal/data-hub/soil-maps-and-databases/en/
217	FAO	soil classification (WRB)	FAO	fao.org/soils-portal/data-hub/soil-classification/en/
218	FAO	soil salinity management	FAO	fao.org/soils-portal/soil-management/management-of-some-problem-soils/

219	FAO	soil organic carbon	FAO	fao.org/global-soil-partnership/areas-of-work/soil-organic-carbon/en/
220	ISRIC	World Soil Information	ISRIC	isric.org
221	ISRIC	SoilGrids global maps	ISRIC	soilgrids.org
222	Soil Science Society of America	glossary	SSSA	soils.org
223	Gypsum for sodic soil remediation	FAO	FAO	fao.org
224	Biochar for soil improvement		IBI	biochar-international.org
225	Compost quality standards		US Composting Council	compostingcouncil.org
226	Soil solarization technique	UC Davis	UC Davis	ipm.ucanr.edu
227	Soil testing and interpretation		FAO	fao.org
228	Sandy soil improvement strategies		FAO	fao.org/soils-portal/en/
229	Calcareous soil management		FAO	fao.org/soils-portal/en/
230	Vermicomposting guide		FAO	fao.org
231	Soil erosion	wind and water	FAO	fao.org/soils-portal/en/
232	Soil biodiversity	FAO	FAO	fao.org/soils-portal/en/
233	Mycorrhizal fungi in agriculture		FAO	fao.org
234	Conservation agriculture principles	FAO	FAO	fao.org/conservation-agriculture/en/

235	No-till farming systems		FAO	fao.org/conservation-agriculture/en/
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B.5 Water & Irrigation

#	Source	Context	Provided by	Link
236	FAO	irrigation and drainage papers	FAO	fao.org/land-water/databases-and-software/en/
237	FAO	crop evapotranspiration (FAO-56) · Irrigation scheduling basis	FAO	fao.org/3/x0490e/x0490e00.htm
238	FAO	water quality for agriculture	FAO	fao.org/3/t0234e/t0234e00.htm
239	FAO	irrigation management	FAO	fao.org/land-water/water/en/
240	IWMI	water management institute	IWMI	iwmi.cgiar.org
241	ICID	International Commission on Irrigation	ICID	icid.org
242	Drip irrigation design and management		FAO	fao.org/land-water/water/en/
243	Subsurface drip irrigation		ASABE	asabe.org
244	Centre-pivot irrigation efficiency		FAO	fao.org
245	Deficit irrigation strategies	FAO	FAO	fao.org/land-water/water/en/
246	Rainwater harvesting	FAO	FAO	fao.org/land-water/water/en/
247	Traditional water systems	aflaj and qanats	UNESCO	whc.unesco.org
248	Desalination for agriculture		ICBA	biosaline.org

249	Solar-powered irrigation		FAO	fao.org/land-water/water/en/
250	Treated wastewater reuse in agriculture	FAO	FAO	fao.org/land-water/water/en/
251	Groundwater management	FAO	FAO	fao.org/land-water/water/en/
252	Water footprint calculator		Water Footprint Network	waterfootprint.org
253	Virtual water trade	WFN	Water Footprint Network	waterfootprint.org
254	Saudi water desalination	SWCC	SWCC	swcc.gov.sa/English
255	Saline Water Conversion Corporation		SWCC	swcc.gov.sa/English
256	Hydroponics	technical guide	FAO	fao.org
257	Aquaponics	FAO technical paper	FAO	fao.org/3/i4021e/i4021e.pdf
258	Greenhouse water management		FAO	fao.org

B.6 Pests, Diseases & Plant Protection

#	Source	Context	Provided by	Link
259	FAO	integrated pest management	FAO	fao.org/pest-and-pesticide-management/ipm/en/
260	FAO	pesticide management	FAO	fao.org/pest-and-pesticide-management/en/
261	FAO	red palm weevil programme	FAO	fao.org/red-palm-

				weevil/en/
262	FAO	desert locust watch	FAO	fao.org/locusts/en/
263	FAO	fall armyworm	FAO	fao.org/fall-armyworm/en/
264	CABI	Crop Protection Compendium	CABI	cabi.org/cpc
265	CABI	Plantwise knowledge bank	CABI	cabi.org
266	EPPO	European plant protection standards	EPPO	eppo.int
267	IPPC	International Plant Protection Convention	FAO/IPPC	ippc.int
268	Biological control of pests	IOBC	IOBC	iobc-global.org
269	Whitefly management in greenhouses		FAO	fao.org
270	Spider mite management		UC IPM	ipm.ucanr.edu
271	Fruit fly management	IAEA	IAEA	iaea.org/topics/sterile-insect-technique
272	Nematode management	UC Davis	UC Davis	ipm.ucanr.edu
273	Anthraxnose management in tropical fruit		CABI	cabi.org
274	Powdery mildew	integrated management	UC IPM	ipm.ucanr.edu
275	Coffee leaf rust	strategies	WCR	worldcoffeeresearch.org
276	Citrus canker management		USDA	aphis.usda.gov
277	Fusarium wilt management		FAO	fao.org
278	Bayoud disease of date palm		FAO	fao.org
279	Pesticide resistance	IRAC	IRAC	irac-online.org

	management			
280	Neem oil as botanical pesticide		FAO	fao.org
281	Bt (Bacillus thuringiensis) biopesticide		WHO	who.int
282	Pheromone traps for pest monitoring		CABI	cabi.org
283	Plant quarantine and biosecurity	IPPC	IPPC	ippc.int

B.7 Climate & Weather

#	Source	Context	Provided by	Link
284	IPCC	climate change and food security	IPCC	ipcc.ch
285	IPCC	land and climate	IPCC	ipcc.ch
286	WMO	World Meteorological Organization	WMO	wmo.int
287	WMO	agrometeorological bulletins	WMO	wmo.int
288	PME	Saudi Presidency of Meteorology	PME	pme.gov.sa/en
289	NASA POWER	solar and weather data Agroclimatology data	NASA	power.larc.nasa.gov
290	ERA5 reanalysis	climate data	ECMWF / Copernicus	cds.climate.copernicus.eu
291	WorldClim	global climate data	WorldClim	worldclim.org
292	Climate-ADAPT	adaptation platform	EEA	climate-adapt.eea.europa.eu
293	CGIAR	climate change and food systems	CGIAR	cgiar.org
294	Global Chill-Hour Calculator		AgWeatherNet	weather.wsu.edu

295	Heat stress in crops	ICRISAT	ICRISAT	icrisat.org
296	Frost protection methods	FAO	FAO	fao.org
297	Shade netting for heat reduction		FAO	fao.org

B.8 Technology & Precision Agriculture

#	Source	Context	Provided by	Link
298	FAO	digital agriculture	FAO	fao.org/digital-agriculture/en/
299	FAO	e-agriculture	FAO	fao.org/e-agriculture/en/
300	Precision agriculture overview	USDA	USDA	usda.gov
301	NDVI	vegetation index applications	NASA	earthobservatory.nasa.gov
302	Sentinel-2	satellite crop monitoring	ESA / Copernicus	sentinel.esa.int/web/sentinel/missions/sentinel-2
303	Google Earth Engine	agriculture	Google	earthengine.google.com
304	Drones in agriculture	FAO	FAO	fao.org/digital-agriculture/en/
305	IoT sensors for irrigation		FAO	fao.org/digital-agriculture/en/
306	AI in agriculture	CGIAR	CGIAR	cgiar.org
307	Machine learning for crop prediction		FAO	fao.org
308	Digital soil mapping		ISRIC	isric.org

309	Farm management information systems		FAO	fao.org
310	Blockchain for food traceability		FAO	fao.org
311	Robotics in agriculture		ASABE	asabe.org
312	Vertical farming technology		Association for Vertical Farming	vertical-farming.net
313	Greenhouse technology	ISHS	ISHS	ishs.org
314	Evaporative cooling for greenhouses		ISHS	ishs.org
315	Solar energy in agriculture		IRENA	irena.org

B.9 Food Security & Agricultural Economics

#	Source	Context	Provided by	Link
316	Global Food Security Index		Economist Impact	impact.economist.com/sustainability/project/food-security-index
317	IFPRI	food policy research	IFPRI	ifpri.org
318	FAO	State of Food Security (SOFI)	FAO	fao.org/publications/sofi/en/
319	WTO	agricultural trade	WTO	wto.org/english/tratop_e/agric_e/agric_e.htm
320	GCC food security strategy		GCC	gcc-sg.org/en-us
321	SALIC	Saudi Agricultural and Livestock Investment	SALIC	salic.com
322	Organic agriculture	IFOAM	IFOAM	ifoam.bio

323	Good Agricultural Practice (GAP)		FAO	fao.org/good-agricultural-practices/en/
324	GlobalG.A.P. certification		GlobalG.A.P.	globalgap.org
325	Post-harvest loss reduction	FAO	FAO	fao.org/food-loss-and-food-waste/en/
326	Cold chain management	FAO	FAO	fao.org
327	Agricultural cooperatives	ICA	ICA	ica.coop
328	Smallholder farmer support	IFAD	IFAD	ifad.org
329	Rural development	World Bank	World Bank	worldbank.org/en/topic/agriculture
330	Agri-tourism development		UNWTO	unwto.org
331	Value chain development in agriculture		FAO	fao.org
332	Agricultural insurance	World Bank	World Bank	worldbank.org

B.10 Livestock, Bees & Aquaculture

#	Source	Context	Provided by	Link
333	FAO	livestock sector	FAO	fao.org/animal-production/en/
334	FAO	dairy production	FAO	fao.org/dairy-production-products/en/
335	FAO	poultry production	FAO	fao.org/poultry-production-products/en/
336	Camel production	FAO	FAO	fao.org

337	Camel milk industry	Saudi Arabia	MEWA	mewa.gov.sa/en/
338	Small ruminant production	ICARDA	ICARDA	icarda.org
339	Rangeland management	FAO	FAO	fao.org
340	Bee health and pollination	FAO	FAO	fao.org/pollination/en/
341	Apimondia	world beekeeping	Apimondia	apimondia.com
342	Varroa mite management		FAO	fao.org
343	FAO	aquaculture	FAO	fao.org/fishery/en/aquaculture
344	Saudi Aquaculture Society		SAS	mewa.gov.sa/en/
345	Shrimp farming	World Aquaculture Society	WAS	was.org
346	Tilapia production guide	WorldFish	WorldFish	worldfishcenter.org
347	Feed alternatives for livestock		FAO	fao.org

B.11 Extension, Education & Additional Resources

#	Source	Context	Provided by	Link
358	FAO	farmer field schools	FAO	fao.org/farmer-field-schools/en/
359	FAO	extension and advisory services	FAO	fao.org/in-action/forest-and-farm-facility/en/
360	GFRAS	global extension forum	GFRAS	g-fras.org
361	UC Cooperative Extension	agriculture	UC ANR	ucanr.edu
362	Texas A&M AgriLife Extension		Texas A&M	agrilifeextension.tamu.edu

363	Arizona Cooperative Extension		U of Arizona	extension.arizona.edu
364	Open access agricultural journals	AGRIS	FAO	agris.fao.org
365	Google Scholar	agricultural research	Google	scholar.google.com
366	PubMed	plant and soil science	NIH	pubmed.ncbi.nlm.nih.gov
367	ResearchGate	agricultural researchers	ResearchGate	researchgate.net
368	Springer	crop science journals	Springer Nature	link.springer.com
369	Elsevier	agricultural water management journal	Elsevier	sciencedirect.com
370	Wiley	plant and soil journals	Wiley	onlinelibrary.wiley.com
371	ISHS	horticultural science	ISHS	ishs.org
372	ASHS	American Society for Horticultural Science	ASHS	ashs.org
373	Crop Science Society of America		CSSA	crops.org
374	International Society of Precision Agriculture		ISPA	ispag.org
375	Arab Center for the Study of Arid Zones (ACSAD)		ACSAD	acsad.org
376	Hima conservation tradition	IUCN	IUCN	iucn.org
377	Terracing heritage	FAO GIAHS	FAO	fao.org/giahs/en/
378	Saudi Heritage Commission		Heritage Commission	heritage.moc.gov.sa/en

B.12 Plant Nutrition & Fertilizer Science

#	Source	Context	Provided by	Link
379	IFA	International Fertilizer Association	IFA	fertilizer.org
380	IPNI	International Plant Nutrition Institute	IPNI	ipni.net
381	4R nutrient stewardship		Fertilizer Canada	fertilizercanada.ca/4rs/
382	Iron deficiency (chlorosis) in calcareous soils		FAO	fao.org/soils-portal/en/
383	Zinc deficiency in crops	FAO	FAO	fao.org
384	Fertigation	nutrient delivery through irrigation	FAO	fao.org
385	Organic fertilizer standards	IFOAM	IFOAM	ifoam.bio
386	Leaf tissue analysis for diagnosis		UC Davis	ucdavis.edu
387	Boron toxicity in arid soils		FAO	fao.org
388	Potassium management in fruit crops		IFA	fertilizer.org
389	Slow-release fertilizers		IFA	fertilizer.org
390	Compost tea	organic nutrient source	USDA	usda.gov
391	Nitrogen fixation by legumes		FAO	fao.org
392	Micronutrient deficiency atlas		HarvestPlus / CGIAR	harvestplus.org
393	Calcium and fruit quality		ISHS	ishs.org

B.13 Post-Harvest Technology & Processing

#	Source	Context	Provided by	Link
394	FAO	post-harvest handling manual	FAO	fao.org/food-loss-and-

				food-waste/en/
395	FAO	cold chain for perishables	FAO	fao.org
396	FAO	drying technologies for fruit	FAO	fao.org
397	Date processing and products	FAO	FAO	fao.org
398	Olive oil extraction technology	IOC	IOC	internationaloliveoil.org
399	Rose water distillation	traditional method	FAO	fao.org
400	Food packaging for tropical climate		FAO	fao.org
401	Solar drying	practical guide	FAO	fao.org
402	Honey extraction and storage		Apimondia	apimondia.com
403	Fruit juice processing		FAO	fao.org
404	HACCP food safety system		FAO / WHO	fao.org/food-safety/food-control-systems/en/
405	Codex Alimentarius	food standards	FAO / WHO	fao.org/fao-who-codexalimentarius/en/
406	Modified atmosphere packaging		FAO	fao.org
407	Ethylene management in storage		UC Davis Postharvest	postharvest.ucdavis.edu
408	Fresh produce grading and sorting		FAO	fao.org

B.14 Greenhouse & Protected Cultivation

#	Source	Context	Provided by	Link
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409	FAO	good agricultural practices in greenhouses	FAO	fao.org
410	Greenhouse crop production	ISHS	ISHS	ishs.org
411	Net house design for hot climates		ICARDA	icarda.org
412	Pad-and-fan cooling systems		ASHRAE	ashrae.org
413	Shade net calibration guide		FAO	fao.org
414	Soilless culture systems	ISHS	ISHS	ishs.org
415	Coco coir as growing medium		FAO	fao.org
416	Rockwool in hydroponics		ISHS	ishs.org
417	Greenhouse insect screening		FAO	fao.org
418	Integrated climate control in greenhouses		Wageningen	wur.nl/en.htm
419	Greenhouse CO2 enrichment		ISHS	ishs.org
420	Seedling nursery management		FAO	fao.org
421	Grafted vegetable seedlings		WorldVeg	avrdc.org
422	Pollination in greenhouses	bumblebees	FAO	fao.org
423	Greenhouse disease management		CABI	cabi.org

B.15 Propagation & Plant Breeding

#	Source	Context	Provided by	Link
424	FAO	plant genetic resources	FAO	fao.org/plant-treaty/en/

425	Svalbard Global Seed Vault		NordGen	seedvault.no
426	GRIN	Germplasm Resources Information Network	USDA	ars-grin.gov
427	Grafting techniques	UC Davis	UC Davis	ucdavis.edu
428	Rootstock selection for salinity tolerance		FAO	fao.org
429	Tissue culture protocols	FAO	FAO	fao.org/biotechnology/en/
430	Seed quality testing	ISTA	ISTA	seedtest.org
431	Plant variety protection	UPOV	UPOV	upov.int
432	Marker-assisted breeding		CGIAR	cgiar.org
433	Heat-tolerant crop varieties	ICRISAT	ICRISAT	icrisat.org
434	Salt-tolerant varieties	ICBA	ICBA	biosaline.org
435	Low-chill fruit breeding programmes		University of Florida	edis.ifas.ufl.edu
436	Drought-tolerant maize	CIMMYT	CIMMYT	cimmyt.org
437	Disease-resistant wheat	ICARDA	ICARDA	icarda.org
438	Date palm breeding	KFU	KFU	kfu.edu.sa/en

B.16 Regional & Historical Agriculture

#	Source	Context	Provided by	Link
449	ACSAD	Arab center for arid zones	ACSAD	acsad.org
450	Arabian Peninsula biodiversity		IUCN	iucn.org
451	GCC-STAT	Gulf statistical centre	GCC-STAT	gccstat.org

452	Kuwait Institute for Scientific Research		KISR	kisir.edu.kw/en/
453	Oman	Ministry of Agriculture	Oman Gov	maf.gov.om
454	UAE	Ministry of Climate Change	UAE Gov	moccae.gov.ae
455	Jordan	agricultural research	NCARE	narc.gov.jo
456	Egypt	Agricultural Research Center	ARC	arc.sci.eg
457	Morocco	agricultural research (INRA)	INRA Morocco	inra.org.ma
458	Tunisia	agricultural research (INRAT)	INRAT	inrat.tn
459	FAO	GIAHS — terracing heritage sites	FAO	fao.org/giahs/en/
460	UNESCO	cultural landscapes of agriculture	UNESCO	whc.unesco.org
461	Traditional irrigation systems	qanat	UNESCO	whc.unesco.org
462	Falaj irrigation	Oman World Heritage	UNESCO	whc.unesco.org/en/list/1207/
463	Al-Ahsa Oasis	World Heritage site	UNESCO	whc.unesco.org/en/list/1563/
464	Hima	traditional conservation system	IUCN	iucn.org
465	NEOM	sustainable food production	NEOM	neom.com
466	The Red Sea Project	marine environment	Red Sea Global	redseaglobal.com
467	Saudi coffee terraces and traditional knowledge		UNESCO ICH	ich.unesco.org
468	Taif rose cultivation heritage		Saudi Tourism	visitsaudi.com

B.17 Tools, Software & Open Data

#	Source	Context	Provided by	Link
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469	QGIS	open-source GIS	QGIS	qgis.org
470	R	statistical computing for agriculture	R Foundation	r-project.org
471	Python	scientific computing	Python	python.org
472	TensorFlow	machine learning	Google	tensorflow.org
473	OpenWeatherMap	weather API	OpenWeather	openweathermap.org
474	Open Data Kit	field data collection	ODK	getodk.org
475	KoboToolbox	survey data	KoboToolbox	kobotoolbox.org
476	CropSAT	satellite crop monitoring	CropSAT	cropsat.com
477	Global Biodiversity Information Facility		GBIF	gbif.org
478	Plants of the World Online	Kew	RBG Kew	powo.science.kew.org
479	Crop Ontology	CGIAR	CGIAR	croponontology.org
480	Agrisemantics	agricultural vocabulary	FAO	aims.fao.org
481	AGROVOC	agricultural thesaurus	FAO	agrovoc.fao.org
482	VS Code	development environment	Microsoft	code.visualstudio.com
483	Live Server	VS Code extension	Ritwick Dey	marketplace.visualstudio.com/items?itemName=ritwickdey.LiveServer
484	Google AI Studio	Gemini API keys	Google	aistudio.google.com
485	GitHub	version control and hosting	GitHub	github.com
486	GitHub Pages	free static hosting	GitHub	pages.github.com
487	MDN Web Docs	web standards	Mozilla	developer.mozilla.org
488	W3C	web accessibility (WCAG)	W3C	w3.org/WAI/

489	Can I Use	browser compatibility	Alexis Deveria	caniuse.com
490	Stack Overflow	developer community	Stack Overflow	stackoverflow.com
491	FAO	women in agriculture	FAO	fao.org/gender/en/
492	IFPRI	gender in agriculture	IFPRI	ifpri.org
497	FAO	State of World's Forests (SOFO)	FAO	fao.org/state-of-forests/en/
498	FAO	global biodiversity for food	FAO	fao.org/biodiversity/en/
499	Saudi Royal Commission for AIUla	heritage agriculture	RCU	rcu.gov.sa/en/
500	Tabuk roses heritage		Saudi Tourism	visitsaudi.com
501	Saudi Arabia country profile	World Bank	World Bank	data.worldbank.org/country/saudi-arabia
502	Circular carbon economy	KSA G20 presidency	G20 Saudi	g20.org
503	World Agroforestry	systems for dry regions	ICRAF	worldagroforestry.org
504	IWA	International Water Association	IWA	iwa-network.org
505	ISSCAS	International Society of Soil Sciences	IUSS	iuss.org

Summary

Part A (Policy, Standards & Governance): 83 sources

Part B (Agricultural Domain & Technical References): 422 sources

Total: 505 sources — unchanged from the original submission, all now clickable.

Spot-checked during review: ITU-T Y.3172, MEWA, and Vision 2030 resolved correctly. SDAIA and GASTAT pages were blocked by automated-fetch protection (common for government WAFs) and should be opened manually once before final submission.